Dean M. Bisogno

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SUMMARY

Data and platform engineer who designs and builds the pipelines, models, and applications that turn messy operational data into production systems. Pairs a Ph.D. in mathematics and peer-reviewed research with hands-on delivery: dbt and Snowflake at enterprise scale, full-stack applications in Python and TypeScript, and a multi-repository data platform designed and built solo for an education non-profit. Six years building production software before graduate school. Seeking deep technical work as a senior individual contributor or hands-on technical lead.

PROFESSIONAL EXPERIENCE

People Analytics Director (analytics and data engineering), JBS USA, Greeley, CO Mar 2022 – Present

- Lead enterprise analytics and data engineering at North American and global scope, serving HR leadership, the CHRO and CFO, Legal, business-unit heads, and the JBS S.A. board across North America, Europe, Australia, and Brazil.
- Contribute roughly 70 commits to the enterprise dbt monorepo on Snowflake across Finance, HR, Sales, Supply Chain, and Operations: rebuilt UKG absenteeism models with unified paycode and attendance-event absences, seed-backed classification, and tests, and eliminated global headcount drift across US/CA/MX by freezing closed fiscal periods and making location lookups deterministic. Layered staging/intermediate/mart, SQLFluff linting, PR review.
- Lead the migration of more than 40 data products from Dataiku onto dbt and Snowflake, sequenced against concurrent UKG and LMS system transitions, after co-leading the original platform selection with IT.
- Design and build an internal HR application platform: a React and TypeScript front end on a FastAPI service, with Entra ID SSO, server-side manager-scope authorization, and Snowflake access through dbt-governed mart contracts. Architecture decision records, a working fake for every external dependency so the stack runs offline, CI in Azure Pipelines.
- Design a global HR star schema and standardize turnover and retention definitions with regional peers worldwide so each new metric follows one reporting standard. Administer Snowflake access, network policies, and authentication, and set governance standards followed outside the analytics function.
- Build and run a production absence-reporting pipeline spanning Fivetran and Python API ingestion, Snowflake staging, dbt transformation, and Qlik reporting, surfacing unclassified absence events through paycode analysis and worked-hours reconciliation of timecard data.
- Help lead the company's AI adoption program: built the agentic-coding business case with the executive team, ran an internal Claude Code workshop, published shared tooling teammates install, and built a cost-attribution ledger loading token usage idempotently into Snowflake with effective-dated pricing behind a Streamlit dashboard.
- Lead and mentor a three-person analytics team, having elevated a team member into the manager role, and advise on technical hiring across business units. Recruited and trained a senior hire to take over compensation program administration, refocusing my own role on analytics and engineering.

Volunteer Data Platform Engineer, Institut Sangalkam, Senegal (education non-profit) Jun 2026 – Present

- Sole engineer and architect of an eight-repository data platform that turns Senegal's national school census into analysis-ready data for a three-person think tank. Govern it with a written decision register and status ledger that bind the repositories together, with explicit data contracts, manifests, and hash-pinned handoffs between layers.
- Built the extraction layer: a Python pipeline that reads proprietary Microsoft OLAP cube files into a SQLite star schema of 25 fact tables (1.95M rows), 53 conformed dimensions, and 18,952 named schools, keeping the Windows-only ADOMD.NET driver stage separate from a portable, stdlib-only build stage joined by a CSV contract.
- Built the conforming layer in dbt and DuckDB, reconciling vocabularies across school years and publishing Parquet with manifests for downstream analysis, on one pinned toolchain whose acceptance test is behavioral: a fresh clone reaches a green build from the README alone.
- Built reference ingestion from World Bank, UNESCO UIS, UN population, and national statistics sources, including a spec-driven reader for regional indicators published only in PDF, gated by validation against external benchmarks and internal consistency before anything publishes. Wrote a separate reader for the one school year that shipped no cubes, documenting its weaker reconciliation guarantee rather than letting it blend in silently.

Data Scientist II, JBS USA, Greeley, CO Feb 2021 – Mar 2022

- Embedded data scientist in industrial workforce planning, independently owning problem identification, data engineering, modeling, and executive presentation of results.
- Designed, built, and deployed VaxTrax, an in-house COVID-19 vaccination tracking application in Python and Flask, from scratch to production to meet an urgent need ahead of OSHA emergency directives.
- Co-led platform evaluation with IT, selecting Snowflake and Dataiku, and built the initial ETL pipelines and Python tooling that remain in production today.
- Deployed XGBoost turnover-risk models, published an analysis linking shift length and task switching to injury rates, and built a graph-theoretic model of viral transmission through processing facilities to forecast policy efficacy. Named 2022 Innovator by the JBS Center for Innovation and Technological Excellence.

Graduate Research & Teaching Assistant, Colorado State University, Fort Collins, CO Aug 2015 – May 2021

- Built research software in Python, Magma, and Gap to automate verification of open problems in algebraic number theory, contributing to two peer-reviewed publications.
- Led a six-person undergraduate research team studying curves amenable to cryptographic applications, mentoring students in theory and scientific programming. Taught Cryptography, Group Theory, Linear Algebra, and Calculus; awarded Outstanding Graduate Teaching Assistant.

Lead Developer, Paladin Innovators, Federal Way, WA Sept 2009 – Aug 2015

- Built early live video streaming platforms at a startup, spanning server infrastructure (C#, Python), native iOS and web clients, and interactive viewer control, filing three provisional patents on systems that anticipated capabilities now standard in cloud-based streaming.

SKILLS

Cloud & Data Platforms: Snowflake, dbt, Fivetran, DuckDB, BigQuery, Dataiku, Parquet, SQLite, MongoDB, Redis, Qlik Sense
Languages & Frameworks: Python, SQL, Pandas, NumPy, FastAPI, Flask, Streamlit, TypeScript, JavaScript, React, Node.js, C#
AI/ML: PyTorch, scikit-learn, XGBoost, Claude Code, Snowflake Cortex, GitHub Copilot
Statistics & Methods: Statistical analysis, regression, gradient-boosted trees, quasi-experimental analysis on observational data, dimensionality reduction, graph modeling
DevOps & Tools: Linux, Docker, Git, CI/CD, Azure DevOps, Azure Pipelines, Entra ID, Windows Server, IIS, Apache, Render

EDUCATION

Ph.D., Mathematics, Colorado State University, Fort Collins, CO 2020
Dissertation: Arithmetic Properties of Curves and Jacobians

M.S., Mathematics, Colorado State University, Fort Collins, CO 2018
Thesis: Abhyankar's Inertia Conjecture for Certain Sporadic Groups

PUBLICATIONS

- Abhyankar's Inertia Conjecture for Some Sporadic Groups. Dean Bisogno. Preprint. arXiv:2010.01455 (2020).
- Group-theoretic Johnson classes and a non-hyperelliptic curve with torsion Ceresa class. Dean Bisogno, Wanlin Li, Daniel Litt, Padmavathi Srinivasan. Épijournal de Géométrie Algébrique, Volume 7 (2023).
- The Supersingularity of Hurwitz Curves. Erin Dawson, Henry Frauenhoff, Michael Lynch, Amethyst Price, Seamus Somerstep, Eric Work, Dean Bisogno, Rachel Pries. Involve, a Journal of Mathematics 12-8 (2019).